Supplementary Materials for

Detecting convergence of amino acid physicochemical property underlying the organismal adaptive convergent evolution

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This PDF file includes:

Figs. S1 to S9

Tables S2, S3, S6, S8-S17

Supplementary References

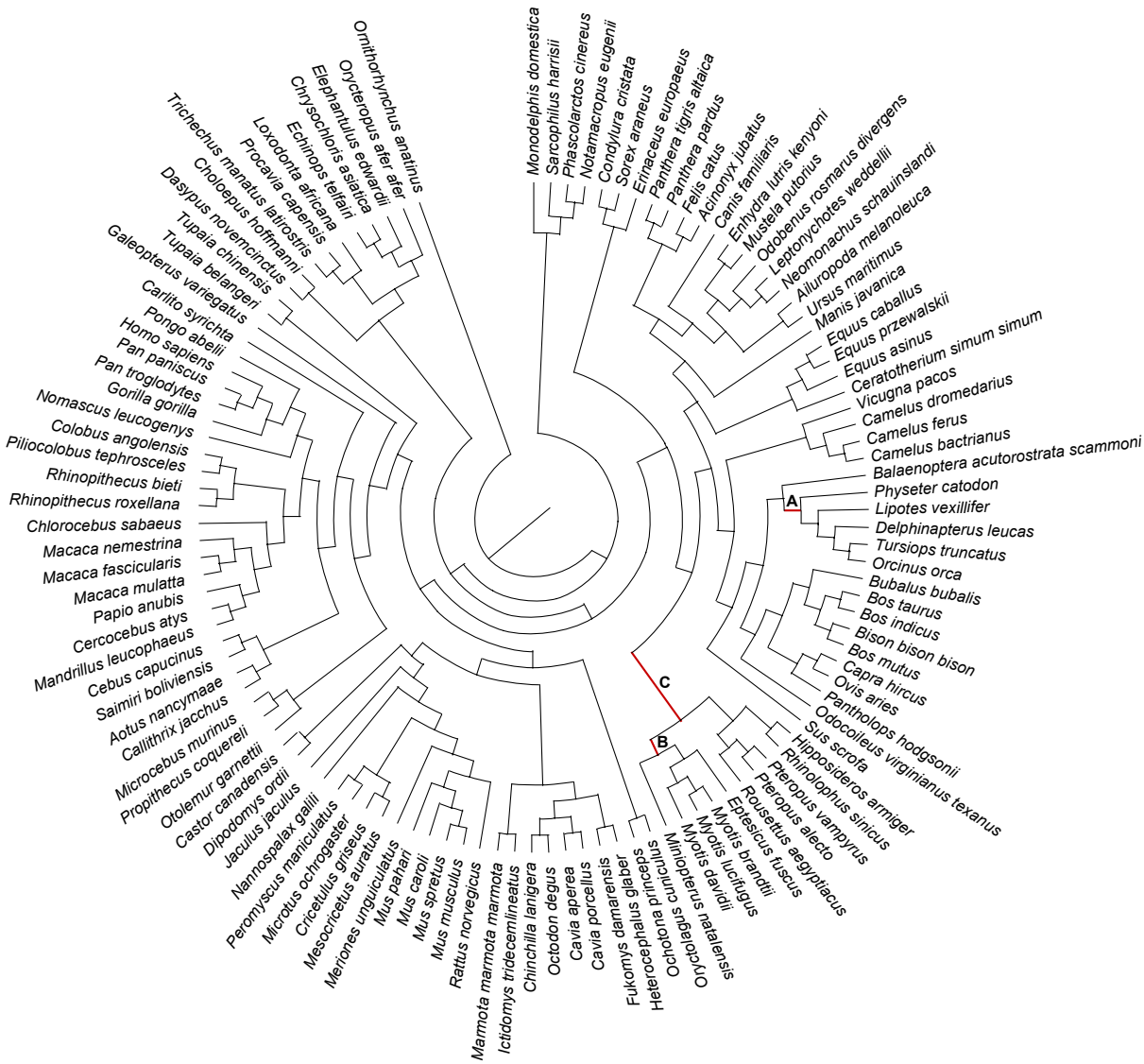

Fig. S1.

Species tree of 116 mammals used in the detection of property convergence.

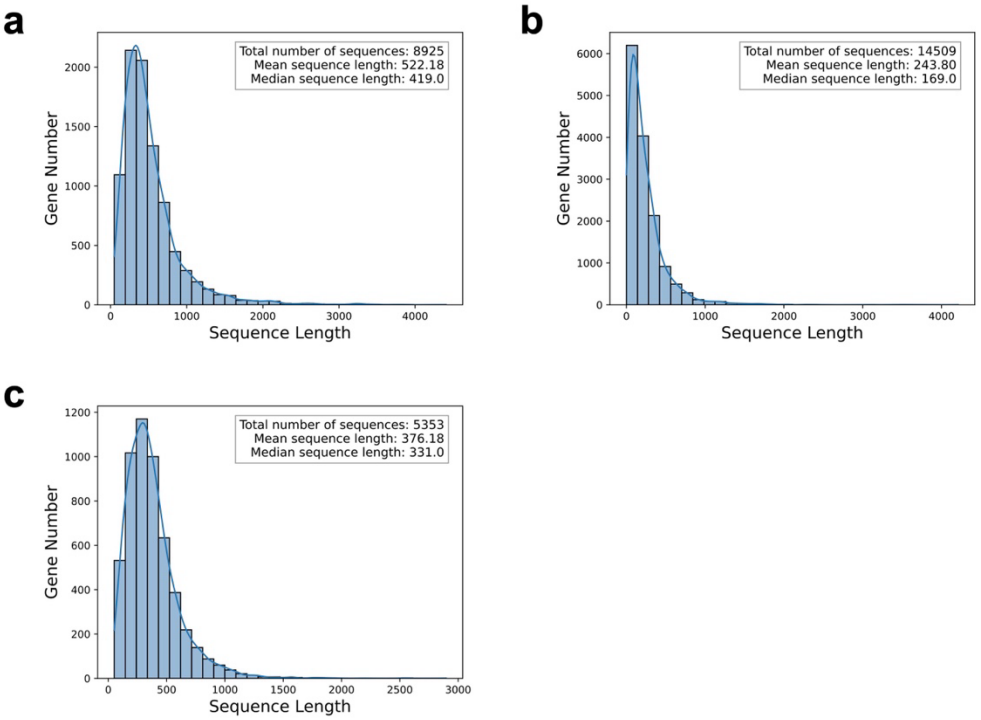

Fig. S2.

Distribution of sequence length in each case. (a) Sequence length distribution of genes used in echolocation case. (b) Sequence length distribution of genes used in marine mammal case. (c) Sequence length distribution of genes used in mangrove case.

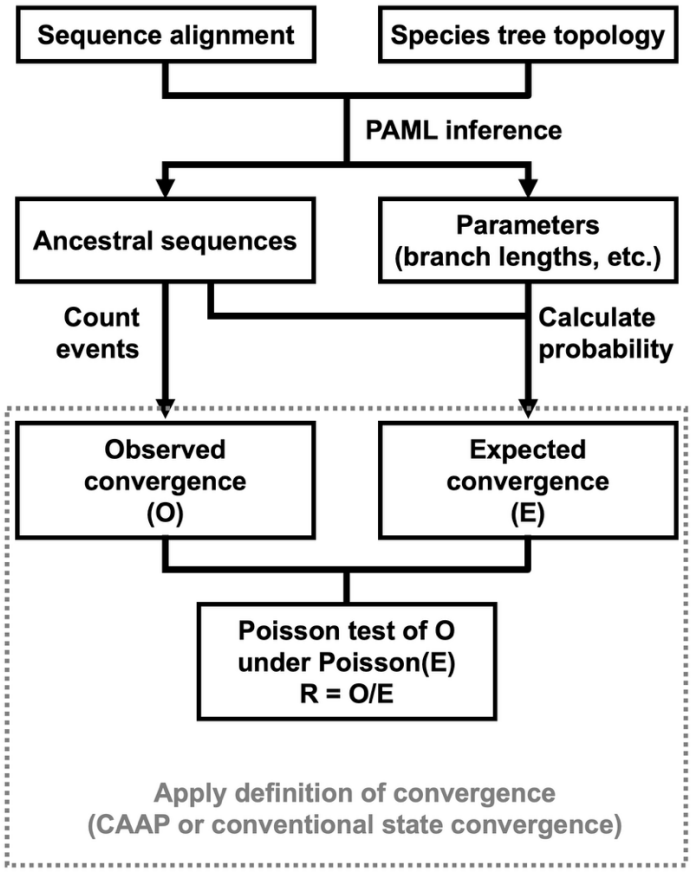

Fig. S3.

Schematic overview of the CAAP gene identification workflow.

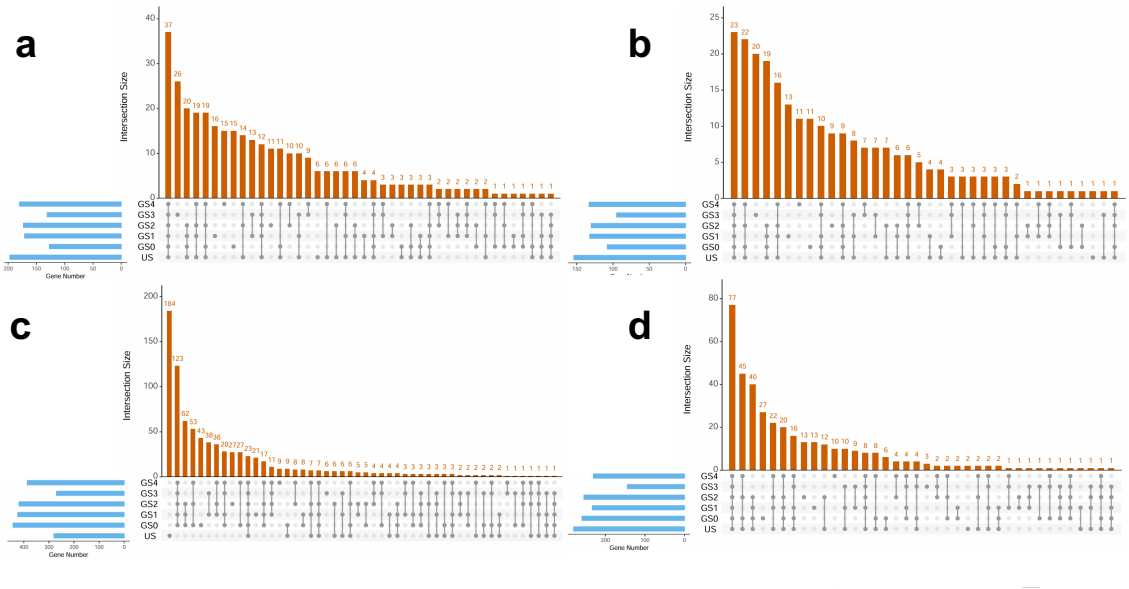

Fig. S4.

Overlap of convergent genes found under different schemes for echolocating mammals. (**a**) Venn diagram of genes with R > 1 between GSs and the US detected in BP1. (**b**) Venn diagram of genes with R > 1 between GSs and the US detected in BP2. (**c**) Venn diagram of genes with O > 0 between GSs detected in BP1. (**d**) Venn diagram of genes with O > 0 between GSs detected in BP2.

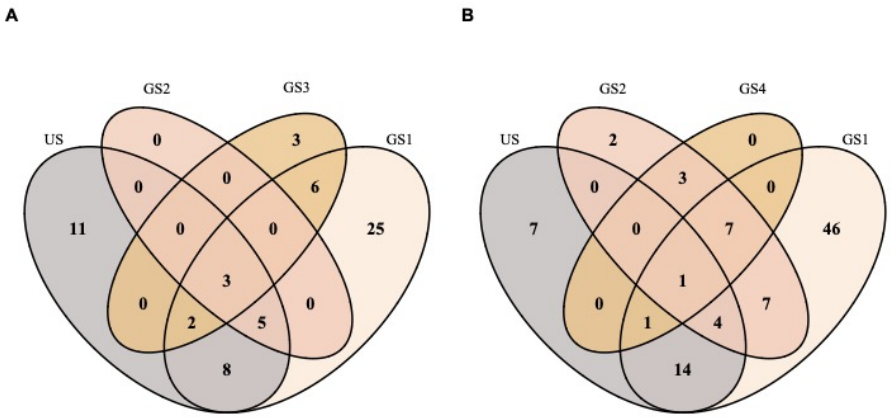

Fig. S5.

Overlap of enriched genes in significant GO terms found by genes with R > 1 in US and GS1-4. (**a**) Venn diagram of enriched genes in US and GS1-4 in BP1. (**b**) Venn diagram of enriched genes in US and GS1-4 in BP2.

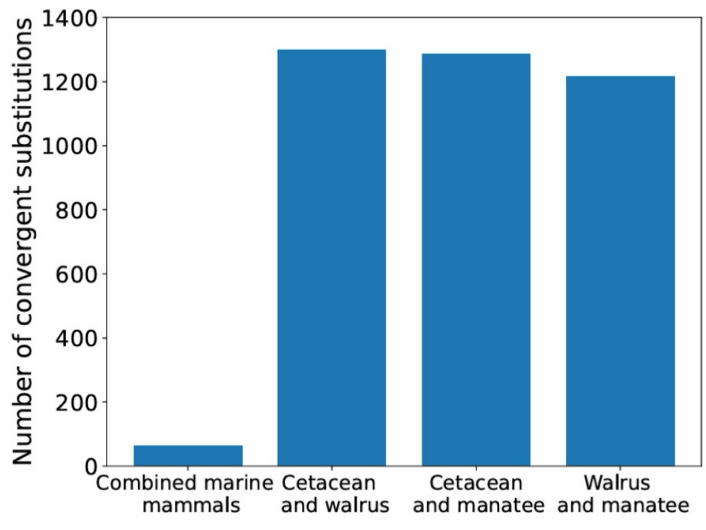

Fig. S6.

Number of convergent sites for each lineage combination, corresponding to Fig. 2 in Foote, et al. (2015).

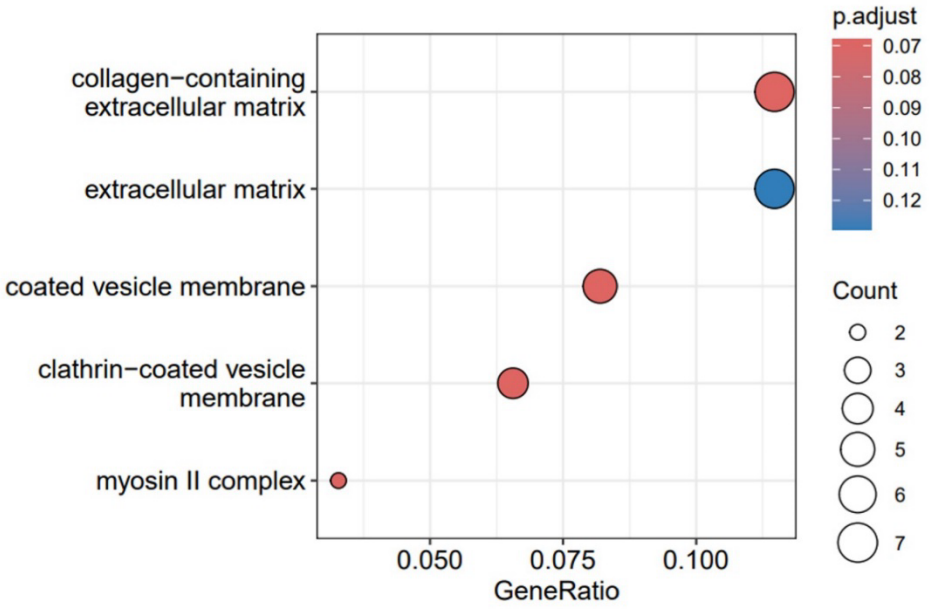

Fig. S7.

GO enrichment result of the gene set with three-lineage convergent sites under US in marine mammals.

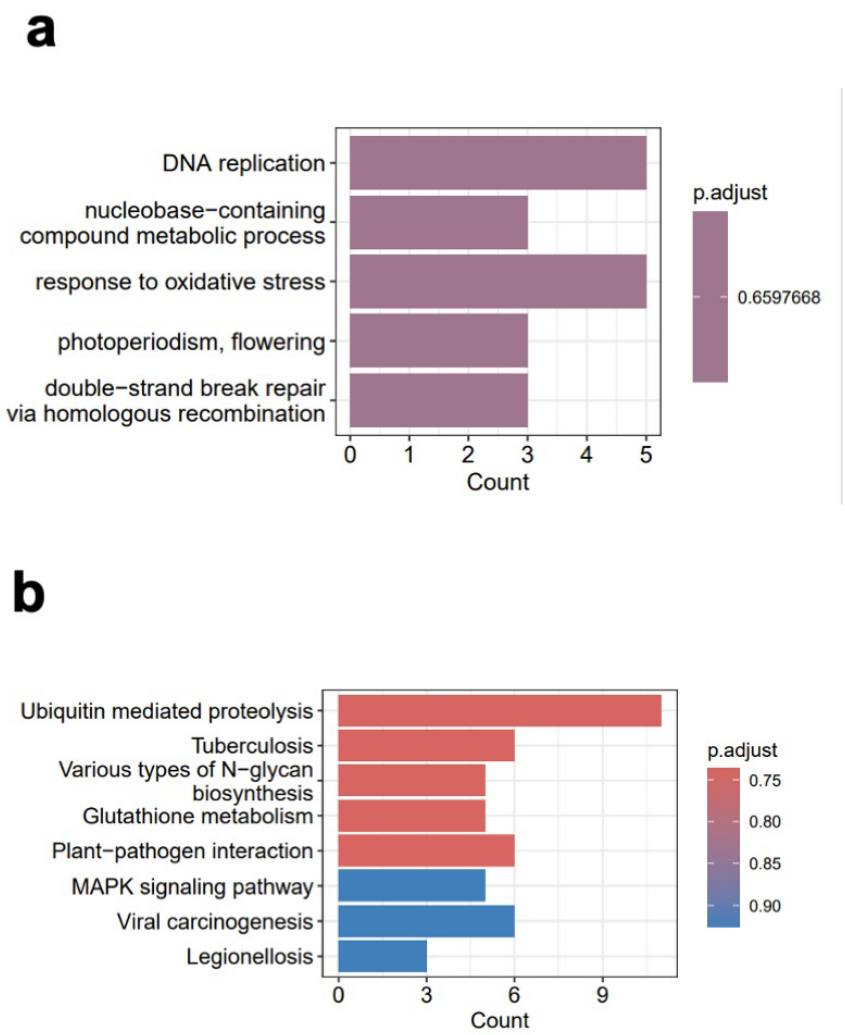

Fig. S8.

Enrichment results of the convergent genes under US in mangroves. (**a**) GO enrichment in mangroves under US. (**b**) KEGG enrichment in mangroves under US.

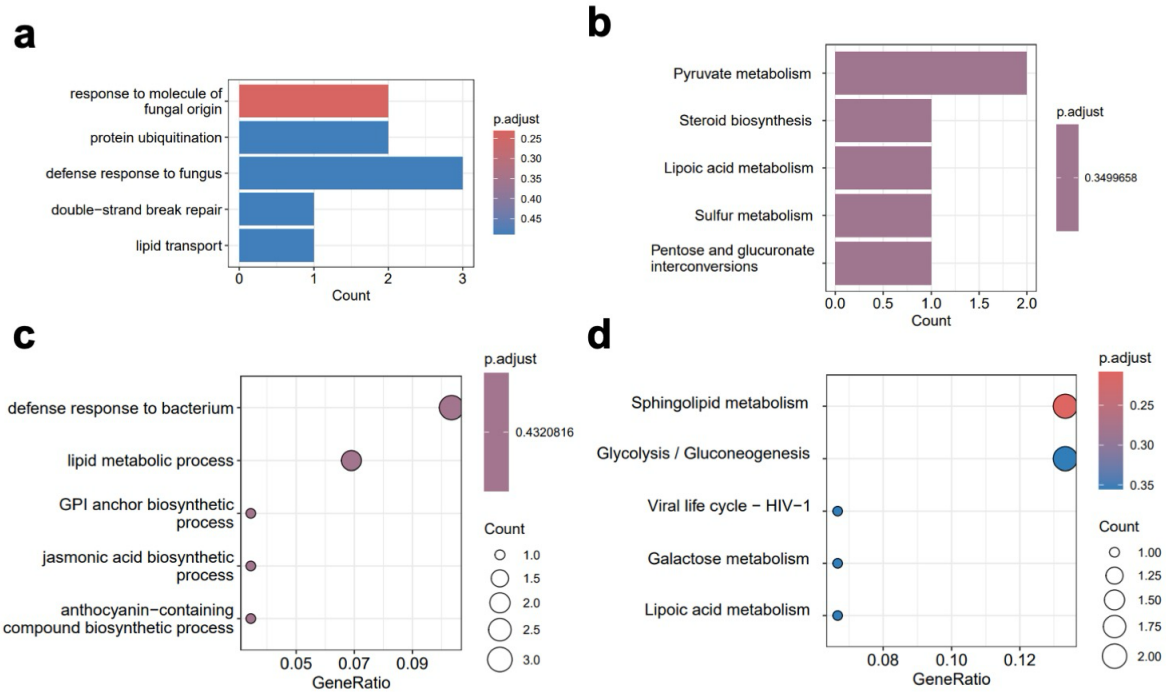

Fig. S9.

Enrichment results of the gene sets with increased convergent sites under GS1 and GS4 in non-mangroves. (**a**) GO enrichment in non-mangroves under GS1. (**b**) KEGG enrichment in non-mangroves under GS1. (**c**) GO enrichment in non-mangroves under GS4. (**d**) KEGG enrichment in non-mangroves under GS4.

Table S2. The O, E, R and Poisson test *P* values for US and all GSs in two lineages of the 1,000 sequences alignments simulated under the neutral substitution models.

Statistic	US	GS0	GS1	GS2	GS3	GS4
O	1788	2739	1584	1926	973	1425
E	1784.509	2717.753	1603.804	1946.654	999.1578	1411.927
R	1.002	1.008	0.988	0.989	0.974	1.009
***P* value**	0.934	0.680	0.635	0.650	0.420	0.719

Table S3. Compare the results of genes with R > 1 and the PCOC model for seven previous genes associated with adaptive functions in echolocating mammals. The values in US and GS0-4 represent the Poisson test *P* values. The *P* values < 0.05 in GS1-4 were highlighted in red. The '-' indicated no CAAP was found, and the Y / N corresponding to genes can / cannot be found by PCOC.

Gene	BP	US	GS0	GS1	GS2	GS3	GS4	PCOC
SLC26A5	BP1	-	-	-	-	-	-	N
	BP2	0.0575	0.0796	-	0.0628	-	0.0297	N
PJVK	BP1	0.0434	0.0567	-	-	-	0.0230	N
	BP2	0.0135	0.0176	0.0122	0.0122	-	-	N
OTOF	BP1	0.00674	0.0132	0.103	-	-	-	N
	BP2	-	-	-	-	-	-	Y
CDH23	BP1	0.0221	-	0.0227	-	0.00738	-	Y
	BP2	-	-	-	-	-	-	Y
TMC1	BP1	0.000147	0.00925	0.00494	0.00629	0.0469	0.0673	Y
	BP2	0.0299	-	0.0312	-	0.0141	-	Y
KCNQ4	BP1	-	-	-	-	-	-	N
	BP2	-	-	-	-	-	-	N
PCDH15	BP1	0.103	0.140	0.100	0.108	-	-	Y
	BP2	-	-	-	-	-	-	N

Table S6. The number of genes with R > 1 between the echolocating bats and non-echolocating bovids under the US and GS1-4.

Branch pair	US	GS1	GS2	GS3	GS4
BP1	328	273	268	218	297
BP2	255	205	226	160	224

Table S8. GO enrichment result of genes with R > 1 (*P* value < 0.05) in all combinations of GS1-4 and BPs for echolocating mammals.

Gene	Amino acid change	Functional relations
GS1, BP1		
GUCY1B1	(200_NNTT)*	Encodes the beta subunit of the soluble guanylate cyclase (sGC), which produces cGMP in central adaptive auditory responses (Calis, et al. 2023).
ADGRF5	(212_NNSS)	-
NECTIN1	(131_TTAS)	Nectin-1 is crucial for maintaining the checkerboard-like cellular pattern in the cochlear auditory epithelium (Katsunuma, et al. 2022).
TRPV4	(563_VVAA)	Encodes a mechanosensitive non-selective cation channel that is essential for maintaining normal hearing thresholds and protecting the cochlea from acoustic injury (Tabuchi, et al. 2005).
PLXNA1	(421_QQRR)	The *PLXNA1* variant can cause variable sensorineural hearing loss (SNHL) (Dworschak, et al. 2021).
PTCH2	(763 RRQQ,822 M MVV)	-
PTPRB	(1011_IIVV)	A patient with deletion of a genomic region including PTPRB suffers from hearing loss (Alesi, et al. 2017).
P2RX4	(158_TTAS)	Encodes a purinergic receptor subunit that mediates cation absorption in utricular cells together with P2RX2, protecting hair cells from damage during intense auditory stimulation (Jeong, et al. 2020).
GABRR1	(14_IIVV)	-
NTRK2	(37_STFI)	Encodes the high-affinity receptor for brain-derived nerve growth factor (BDNF), which modulates synaptic transmission during the activated status of the auditory system (Gross, et al. 2023).
ADGRF1	(269 NNSS,278 A ATT)	-
KIT	(840_AATT)	Encodes a receptor tyrosine kinase, mutations of KIT ligand (KITLG) can cause asymmetric and unilateral hearing loss (Zazo Seco, et al. 2015).
CHRNB2	(292_VVIL)	Encodes a subunit of neuronal acetylcholine receptor with greater than five-fold differential expression in hair cell-enriched compared to the supporting cells (Cristobal, et al. 2005).
ANTXR1	(319_VVLM)	Encodes Anthrax toxin receptor 1, and mutations in *ANTXR1* cause GAPO syndrome, which includes features such as growth retardation, alopecia, pseudoanodontia, and hearing loss (Hamdi, et al. 2024).
GPR160	(193_IIVV)	-
OR51E1	(45_RRWW)	-
LRRC19	(282_GGRK)	-
GRIN2B	(1119_TTSS)	-
TRPM3	(1315_TTAA)	Encodes a transient receptor potential channel that regulates inner ear fluid homeostasis by sensing osmolality changes, thereby maintaining the function of cochlear and vestibular hair cells and protecting against hearing disorders (Takumida, et al. 2009).
PIEZO2	(617_VVAG)	Encodes a mechanosensitive ion channel essential for ultrasonic hearing via cochlear outer hair cells in mice (Li, et al. 2021).
KCNAB3	(92_AATT)	Regulatory subunit of the voltage-gated potassium (Kv) channels in the cochlear nucleus (Friedland, et al. 2007).
LRRC8E	(647_HHNQ)	Encodes a subunit of the volume-regulated anion channel (VRAC) that is essential for maintaining ion homeostasis in the cochlea, and its disruption leads to cochlear degeneration and congenital deafness (Knecht, et al. 2024).

CLCN1	(47_KKTT,109_TTSS)	-
SLC30A3	(185_PPHH)	Encodes ZnT3, a vesicular zinc transporter for loading zinc into vesicles, and is expressed in cochlear hair cells and the spiral limbus (Bizup, et al. 2024).
SLC27A1	(98_AAVV)	Candidate deafness-associated gene in humans (Uetsuka, et al. 2015).
SLC5A6	(552_LLVV)	-
CNNM4	(337_SSNN)	The encoded protein may play a role in metal ion transport and mutations on *CNNM4* cause recessive cone-rod dystrophy (Polok, et al. 2009).
SLC22A4	(15_PPLL)	Encodes an organic cation transporter expressed in the cochlear strial endothelium, essential for maintaining ion homeostasis, and mutations in *SLC22A4* cause recessive non-syndromic hearing loss DFNB60 (Ben Said, et al. 2016).
GS1, BP2		
PIK3CG	(19_TTAS)	*PIK3CG* was up-regulated when knock-down *GATA3*, which plays a pivotal role in hearing (Luo, et al. 2013). Subunit of PI3K, which is involved in age-related hearing loss (Garcia-Mato, et al. 2021).
GPR17	(1_TTMM)	-
NET1	(186_HHQQ)	Overexpression of *NET1* promoted cochlear organoid formation and hair cell differentiation (Zhang, et al. 2023).
PRAG1	(1096_VVAA)	-
ATP8B1	(815_IIVV)	Encodes the phosphatidylserine flippase ATP8B1, specifically localized in the stereocilia of cochlear hair cells and essential for their integrity and mechanosensory function, with mutations causing progressive hearing loss through hair cell degeneration (Stapelbroek, et al. 2009).
ABCC6	(446_GGEE,1084_VVLL)	Mutations on *ABCC6* are associated with retinal alteration (Tachikawa, et al. 2008; Martin, et al. 2020).
GJB6	(99_HHQQ)	Encode gap junction protein connexin30, which is expressed in mesenchymal structures of the inner ear (Teubner, et al. 2003). Deficiency of connexin30 caused severe hearing impairment in mice (Teubner, et al. 2003).
AOC2	(259_IIVV)	*AOC2* is presented on the cell surfaces of many tissues, with notably high expression in retina (Horn and Wittung-Stafshede 2021).
ZFHX2	(1011_TTSA,1015_HHQQ)	-
SCNN1G	(491_VVII)	Encodes a subunit of ENaC which essential for maintaining a fluid-free middle ear cavity (Choi, et al. 2006).
SCN11A	(738_IIVV,905_GGRR,990_PPLL,1145_EEKK)	Encodes voltage-gated sodium channels, essential for the development of neurons, deletion of which cause sensorineural hearing loss(Zu, et al. 2021).
SLC17A8	(349_VVLL)	Encodes the vesicular glutamate transporter 3 (VGLUT3) specifically expressed in inner hair cells, essential for neurotransmitter transport and synaptic transmission, with mutations causing autosomal dominant non-syndromic hearing loss DFNA25 (Ryu, et al. 2016).
ABCA4	(979_QNHH,1052_KKEQ)	Mutations on *ABCA4* caused retinal disease (Molday 2007; Cremers, et al. 2020).
PJVK	(99_IIVV)	Required for the proper function of auditory pathway neurons and protects and is essential for the development of auditory hair cells (Defourny, et al. 2019).
GPR179	(750_RRQQ,822_GGRR,1336_LLSS)	Encodes an orphan G protein receptor essential for depolarizing bipolar cell signal transduction in the retina, with mutations

		causing autosomal-recessive congenital stationary night blindness (Peachey, et al. 2012).
STXBP3	(148_IIVV)	Expressed in expressed in epithelial, stromal cells and plays crucial role in sensorineural hearing loss (Ouahed, et al. 2021).
DNASE1L3	(242_RKSG)	-
DDX11	(590_TTMM)	Consistently associated with sensorineural or mixed hearing loss in previous studies (Costales, et al. 2020).
UTP15	(10_TTAA)	Expressed in inner hair cells differentially and under positive selection (Pisciottano, et al. 2019).
ENAM	(24_VVII,179_NNSS)	-
MMP20	(123_VVII)	-
VPS13B	(398_VVII,465_QQHH,1552_TTMM)	-
CDHR1	(37_TTAS,75_VVII)	Mutations on *CDHR1* associated with retinal disease (Duncan, et al. 2012).
MTM1	(135_TTMM)	-
CUL7	(559_QQRR)	Mutations on *CUL7* cause 3M syndrome an autosomal recessive disorder with phenotype of hearing loss (Akalın, et al. 2023).
FAT1	(154_VVII)	*FAT* is a member of atypical cadherins (Vanniya. S, et al. 2018). It involves a pathway called DCHS1-FAT4 signalling pathway which is essential for the development of ear and cochlea (Alaqeel, et al. 2019).
PTPRZ1	(667_HYND)	-
RNF6	(468_RQWW)	-
HECW2	(73_NNSS)	-
TWF2	(173_MMTT)	Encodes a protein that regulates the length of stereocilia during early hair bundle development and is particularly enriched in shorter transducing stereocilia (Carlton, et al. 2021).
NFKBIB	(212_RRWW)	Encodes the inhibitor of the NF-κB signaling pathway, which is up-regulated with age and hearing loss (Tadros, et al. 2008).
SLC20A2	(514_IIVV)	-
MAP7	(565_VVAA)	Encodes a microtubule-associated protein that is expressed in multiple cochlear cell types, particularly in outer hair cells and outer pillar cells, with its expression increasing over time (Lewis, et al. 2022).
HPGD	(25_VVLM)	-
LLGL2	(565_SSNN,763_VVII)	Involved in the signaling pathway and increased Ca^{2+} influx (Leng, et al. 2021).
MYH7B	(401_TTAA)	Encodes an evolutionarily conserved myosin heavy chain expressed in nonmuscle tissues, with mutations linked to hereditary hearing loss (Lee, et al. 2023).
GAS2L1	(162_RRQQ)	-
GAS2L2	(465_TTAP)	Associated with childhood ear infection (Bizaki-Vallaskangas, et al. 2024).
POLD1	(186_TTAA)	Encodes DNA polymerase delta 1, and mutations in *POLD1* can cause deafness .(Zuo, et al. 2022)
TOP3B	(704_IIVV)	-
MCM3	(615_RRQQ)	Highly expressed in Lgr5 cochlear hair cell progenitors, indicating a crucial role in their proliferation and maintenance (Zhang, et al. 2018).
ASCC3	(988_IIVV)	-
POLN	(108_MMTT,333_NNSS)	-
LONP1	(779_VVII)	-
SMC5	(746_NNSS)	-

THRAP3	(274_TTAA)	Patient with mutation in *THRAP3* suffer moderate hearing loss (Clabout, et al. 2023).
HDAC7	(296_VVII)	-
MEF2D	(349_QQPP)	Expressed in the developing inner ear (Sanchez-Calderon, et al. 2010).
ANKRD42	(67_VVAA)	-
CACNB1	(204_QQHH)	-
TRPM6	(783_IITT)	Encodes a transient receptor potential channel highly expressed in the stria vascularis, spiral ligament and prominence, outer and inner hair cells, cochlear supporting cells (Takumida, et al. 2009).
GS2, BP2		
SBNO2	(52_TTAA,515_HHRR)	Up-regulated gene in rat cochleae after noise exposure (Yang, et al. 2016).
ITFG2	(374_RRWQ)	-
GS1, BP1 | GS1, BP2		
PKD1	(209_VVMM,797_VVTT,1512_VVII,2295_AATT), (545_VVMM,729_AATT,1749_AAVV,2607_AAVV)	Encodes polycystin-1, a large transmembrane protein essential for maintaining the structure and integrity of stereocilia in cochlear hair cells (Steigelman, et al. 2011).
ABCC3	(546_QQKK,924_IIVV,748_HMQQ), (626_IIVV,698_AAVV)	-
GS1, BP1 | GS3, BP1		
CDK13	(371_TTII)	Encodes a member of the cyclin-dependent kinase family, which reported associated with sensorineural hearing impairment (Hamilton and Suri 2019).
MOK	(209_EEGA,285_RRSS)	-
CDKL1	(319_CCYY)	-
GS1, BP2 | GS2, BP2		
LIPA	(211_IIVV)	-
AXL	(251_GGRR)	-
SCNN1B	(321_MMVV)	Encodes a subunit of ENaC which essential for maintaining a fluid-free middle ear cavity (Choi, et al. 2006).
ADAM11	(79_VVMM)	Encodes a disintegrin and metalloproteinase family protein that is exclusively expressed in amphora-shaped type I hair cells, playing a crucial role in the structural integrity and function of these auditory hair cells (Wang, Ling, et al. 2024).
FYB1	(254_VVMM)	-
LAMA5	(534_TTAA,1020_IIVV,1849_AATT,2112_NNSS,2155_QQRR,2475_RRLL)	Encodes one of the vertebrate laminin alpha chains, whose expression profile changed dynamically in the cochlear epithelial structures (Rodgers, et al. 2001).
LAMC1	(98_IIVV)	Associated with retinal development (Edwards, et al. 2010).
FLNA	(723_IIVV)	Encodes filamin A, and is associated with sensorineural hearing loss (Santos-Cortez, et al. 2021).
CHD7	(1854_TTAA)	Encodes a chromatin remodeler expressed in the endolymphatic sac and duct of the inner ear, with mutations contributing to nonsyndromic hearing loss (Roux, et al. 2023).

SLC39A3	(132_AATT)	-
GS1, BP2 | GS4, BP2		
WISP2	(179_TTMM)	Encodes a member of the Wnt signaling pathway that is significantly upregulated in cochleae three days post-implantation (Bas, et al. 2020).
GS2, BP2 | GS4, BP2		
CD74	(81_PPSS)	-
CBFA2T3	(393_GGSS,513_AAST), (276_MMVL,393_GGSS,513_AAST)	-
CLEC1B	(40_MMIL)	-
GS1, BP1 | GS1, BP2 | GS2, BP2		
ERBB2	(275_SSRR), (143_EEKK)	-
GS1, BP2 | GS2, BP2 | GS3, BP2		
KITLG	(178_SSNN)	Mutations of *KITLG* lead to asymmetric and unilateral hearing loss (Zazo Seco, et al. 2015).
GATA1	(336_VVMM)	A paralog of *GATA2* and *GATA3*, which play pivotal roles in hearing (Holley, et al. 2007; Luo, et al. 2013).
EPB42	(278_NNSS)	-
ZBTB7A	(276_AATT)	-
PRKDC	(334_CCGS,691_RQPP,2424_RGEE)	-
MYH9	(1135_AATT)	Encodes a member of myosin family and defect of *MYH9* have been associated with sensorineural hearing loss (Canzi, et al. 2016). Localized to the stereocilia on the inner and outer hair cells in the inner ear in mice (Friedman, et al. 2020).
ITGB6	(3_VVMM)	Encode a subunit beta 6 of integrin, expressed in the mouse developing inner ear and deficient of which cause Stereocilia defects of inner ear (Littlewood Evans and Müller 2000).
MYC	(195_EEAA)	Encodes a transcription factor involved in regulating cell proliferation and differentiation during inner ear development (Domínguez-Frutos, et al. 2011).

* Content within each bracket indicate a convergence event. The number before the underline indicates the site position in the protein sequence alignment. The four amino acids following the underline denote the two amino acid states at the beginning of the two focal branches, and the two amino acid states at the end of the two focal branches.

Table S9. Genes with R > 1 (*P* value < 0.05) in US and GS1-4, and under positive selection in echolocating mammals.

Gene	Grouping scheme	Branch group	Amino acid change	Functional relations
CCDC92	US | GS4	1	(3_TTAA, 108_KKRR)*	*Ccdc92* knockout mice have abnormal auditory brainstem response that might be compatible with a dysfunction of sensory cilia in the cochlea (Duek, et al. 2018).
CDH19	US|GS2|GS4	1	(360_IITT, 371_NNDD)	*CDH19* expressed in cochlea and significantly down-regulated in a middle ear disease (Klenke, et al. 2012; Lin, et al. 2012).
TEX14	US | GS4	1	(77 RRQQ, 452 RRQQ, 756_AATT, 763_DDAA, 912_RRQQ)	*TEX14* has been reported to show 14.5 times higher expression in the cochlea of mice with impending hearing loss than that of wildtype mice, potentially mediating cellular protrusion formation and cytoskeletal organization (Dufek, et al. 2020).
	GS1	1	(77 RRQQ, 292 VVIL, 452 RRQQ, 756 AATT, 763 DDAA, 912 RRQQ)	
	GS2	1	(77_RRQQ, 292_VVIL, 452_RRQQ, 756_AATT, 763 DDAA, 912 RRQQ, 914 TTIP)	
	GS3	1	(77_RRQQ, 452_RRQQ, 763 DDAA, 912 RRQQ)	
CPA3	US	1	(242 RRGG,294 DDEE, 364_PPSS)	*CPA3* is significantly downregulated in hippocampus after acquired hearing loss (Zhou, et al. 2023).
HPGD	US|GS4	1	(66_RRKK)	*HPGD* encodes 15-hydroxyprostaglandin dehydrogenase and is responsible for metabolism of prostaglandins, which in turn affects cochlea function (Nakagawa 2011; Sun, et al. 2025).
WNK3	GS3	2	(26_PPRH)	*WNK3* can dynamically regulate balance between Na-K-2Cl cotransporters (NKCCs) and K-Cl cotransporters (KCCs) activities (Kahle, et al. 2010). Studies have found that inactivation of NKCCs were associated to deafness (Delpire, et al. 1999), and KCC4 expression can be located in the inner ear and loss for KCC3/KCC4 caused deafness (Boettger, et al. 2002; Boettger, et al. 2003).
JPH3	GS4	2	(403_TTAI)	Increased expression during the maturation of mouse primary auditory cortex (Hackett, et al. 2015)
VCAM1	US	2	(593_LLII)	Encodes a cell surface adhesion molecule involved in endothelial function, with elevated levels in sudden sensorineural hearing loss

				patients (Domínguez-Frutos, et al. 2011).
CHAT	US|GS1|GS2|GS3|GS4	2	(114_PPQQ)	Encodes an enzyme responsible for the synthesis of the neurotransmitter acetylcholine, which is significantly decreased in the auditory cortex of *ADNP* mutant mice (Hacohen-Kleiman, et al. 2019).
ELK3	US|GS1|GS2|GS3|GS4	2	(163_QQPP)	-
CYSTM1	US|GS1|GS2|GS4	2	(79_TTPP)	*CYSTM1* is uniquely expressed in the mature basal cell type of inner ear stria vascularis, which is essential to hearing (Thulasiram, et al. 2025).
PRR14L	US|GS1|GS2|GS3|GS4	2	(1076_SSNN)	*PRR14L* is in a deletion that can cause human hearing loss (Trizuljak, et al. 2023).

* Content within each bracket indicates a convergence event. The number before the underline indicates the site position in the protein sequence alignment. The four amino acids following the underline denote the two amino acid states at the beginning of the two focal branches, and the two amino acid states at the end of the two focal branches.

Table S10. The number of sites showing convergence between echolocating bats and non-echolocating bovids, within genes under positive selection in echolocating mammals.

Gene name	Branch pair	Number of convergent sites				
		US	GS1	GS2	GS3	GS4
CCDC92	BP1	0	0	0	0	0
CDH19	BP1	0	0	0	0	0
TEX14	BP1	3	3	3	2	3
CPA3	BP1	0	0	0	0	0
HPGD	BP1	1	0	0	0	1
WNK3	BP2	-	0	-	0	-
JPH3	BP2	0	0	0	0	0
VCAM1	BP2	0	0	0	0	0
CHAT	BP2	0	0	0	0	0
ELK3	BP2	0	0	0	0	0
CYSTM1	BP2	0	0	0	0	0
PRR14L	BP2	0	0	0	0	0

* A dash ('-') indicated that the R-value for the gene was not significant under the respective grouping scheme.

Table S11. Genes with exclusive property convergence and under positive selection in marine mammals.

Gene	*P* value	FDR	Amino acid change	Functional relations
RIBC1	0.016	0.53	(92_VVVVMMLI)*	Though the function of *RIBC1* remains unknown, its important paralog *RIBC2* plays a significant role in the functioning of motile cilia. Respiration system of marine mammals, which contains motile cilia, is quite different from other terrestrial mammals for exchanging of gas efficiently (Kooyman 1973). Enrich in respiratory ciliated cells according to Human Protein Atlas (proteinatlas.org) (Uhlen, et al. 2010).
SWAP70	0.028	0.68	(291_IIMMVVAV)	May influence muscle morphology by regulating the actin cytoskeleton. Related to coronary artery disease, lung immune cell related, and has lung eQTL (Bahaie, et al. 2012; Nikpay, et al. 2015; Qian, et al. 2023).

* Content within each bracket indicate a convergence event. The number before the underline indicates the site position in the protein sequence alignment. The eight amino acids following the underline denote the four amino acid states at the ancestor nodes of the four marine mammals (the ancestral nodes of the two cetaceans are the same), and the four amino acid states of the four marine mammals.

Table S12. Genes with CAAP enriched in "collagen-containing extracellular matrix" for different GSs in marine mammals.

Gene	Amino acid change	Functional relations
GS1		
ITIH1	(703_VVVVIIMM)*	Expressed in response to lung injury and promote angiogenesis during cystic fibrosis (Garantziotis, et al. 2008).
GS2		
LTBP1	(903_RRRRQQHQ)	Component of extracellular matrix in mouse lung, airway and aorta (Zilberberg, et al. 2012). Influence the formation of elastic fiber and thus may involve in the reconstruction of the respiration system of marine mammals (Jiang, et al. 2016).
COL16A1	(154_VVVASSMT)	Associated with cartilage development in chicken (Takács, et al. 2023).
TINAG	(95_RRRRHHQH)	Component of the core matrisome in lung ECM (Zhou, et al. 2018).
S100A8	(8_IIIIMMMM)	Form heterodimer with *S100A9* and involved in bone and cartilage development in mice (Zreiqat, et al. 2007).
GS1 | GS2		
MMP2	(320_VVVVIIII)	Knockout of *MMP2* cause mice to show asthma with allergen expose (Corry, et al. 2002).
GS1 | GS3		
SPARC	(228_NNNNKKHH)	Associated with lung cancer and pulmonary fibrosis (Wong and Sukkar 2017). Encode osteonectin, which is a bone-specific protein, thus may play role in the bone system development (Termine, et al. 1981)
LTBP2	(1137_MMMMTTTT)	Associated with lung elastinogenesis and fibrosis (Enomoto, et al. 2018).
GS1|GS2|GS3		
LAMA1	(205_QQQQRRRR)	Demonstrated to significantly affect response to lung injury and susceptibility to fibrosis (Lee, et al. 2018).
ERBIN	(438 RRRRQQQQ)	-
SERPINC1	(200_NNNNSSSS)	Regulate blood coagulation; may be associated with low flow rate of viscous blood during diving (Foote, et al. 2015).
ACAN	(357_NNNNSSSS)	Encodes aggrecan, a major component of extracellular matrix essential for the biomechanical properties of cartilage (Shi, et al. 2019).

* Content within each bracket indicate a convergence event. The number before the underline indicates the site position in the protein sequence alignment. The eight amino acids following the underline denote the four amino acid states at the ancestor nodes of the four marine mammals (the ancestral nodes of the two cetaceans are the same), and the four amino acid states of the four marine mammals.

Table S13. The number of genes with CAAP and enriched GO terms in the terrestrial sister species of marine mammals. The '-' means no enriched GO terms or enriched genes were found.

Group scheme	Number of genes with CAAP	Enriched GO terms	Number of enriched genes	FDR
US	108	-	-	-
GS1	112	aminoacyl-tRNA ligase activity	4	0.041425
		ligase activity, forming carbon-oxygen bonds	4	0.041425
GS2	109	-	-	-
GS3	64	-	-	-
GS4	96	-	-	-

Table S14. Non-repetitive enriched GO terms and involved genes for GS0 in marine mammals.

GO term	FDR	Enriched genes	Functional relations
Structural molecule activity	2.40×10^{-5}	*NDC1*	-
		SPTB	-
		LAMA1	Demonstrated to significantly affect response to lung injury and susceptibility to fibrosis (Lee, et al. 2018).
		OLFM4	
		ERBIN	
		LTBP2	Associated with lung elastinogenesis and fibrosis (Enomoto, et al. 2018).
		ACAN	Encodes aggrecan, a major component of extracellular matrix essential for the biomechanical properties of cartilage (Shi, et al. 2019).
		MYL3	-
		KRT19	-
		UMODL1	-
		SYNM	-
Antiporter activity	7.60×10^{-4}	*SLC4A4*	-
		CLCN7	-
		SLC26A1	-
		SLC9B1	-

Table S15. Number of convergent genes for mangroves and non-mangrove and the result of KEGG (corresponding to the Fig. 2a in Xu, et al. (2017))

Pathway	Total # of genes in the category	# of convergent genes		*P* value
		mangroves	Non-mangroves	
All	5353	232	117	-
Ubiquitin mediated proteolysis	59	11	0	0.0111 (FDR = 0.737)
Various types of N-glycan biosynthesis	23	5	1	0.0161 (FDR = 0.737)

Table S16. All significantly enriched GO terms under different GS in mangroves.

Grouping scheme	GO term	Ontology class	Number of enriched genes	FDR
GS4	transferase activity, transferring phosphorus-containing groups	MF	13	0.006
GS4	protein kinase activity	MF	12	0.009
GS4	ATP binding	MF	21	0.018
GS4	protein serine/threonine kinase activity	MF	9	0.044
GS4	protein tyrosine kinase activity	MF	6	0.044
GS4	Protein phosphorylation	BP	12	0.041
GS1	ribonucleoside triphosphate phosphatase activity	MF	7	0.030

Table S17. GO and KEGG pathway enrichment result of the gene sets with increased convergent sites in mangroves under GS1 and GS4.

Gene symbol/MSU	Amino acid change	Functional relations
protein phosphorylation | transferase activity, transferring phosphorus-containing groups | ATP binding (GO term)		
WNK1	(113_QEEQQQQ,238_ISTIIII)*	Transgene of soybean *WNK1* into Arabidopsis significantly enhanced tolerance to NaCl and osmotic stresses (Wang, et al. 2011). Study have found that *WNK1* was regulated by abiotic stress such as salt, heat and involved in maintaining rhythmic behavior in rice (Kumar, et al. 2011).
SPARK2	(29 IFFIIII,427 SRSRRRR, 428_QSNSSSS,436_GEGEEEE)	Downregulated under salt stress in rice (Wang, et al. 2022).
RLCK69	(91_IATTTTT,202_LLMMMMM, 248_EQQEEEE,285_LSSLLLL)	Members of Receptor-like cytoplasmic kinases (RLCK) gene family which play roles in stress response in plants (Vij, et al. 2008).
RLCK172	(36_RKKRRRR,45_SQNSSSS, 47_RKRKKKK,296_PPSSSSS)	Members of Receptor-like cytoplasmic kinases (RLCK) gene family which play roles in stress response in plants (Vij, et al. 2008).
RLCK212	(3_NEQEEEE,190_YYFFFFF, 193_LHLHHHH)	Members of Receptor-like cytoplasmic kinases (RLCK) gene family which play roles in stress response in plants (Vij, et al. 2008).
CDKE1	(394 VIIMMMM,415 PSSPPPP, 420_IMLMMMM)	Mediate the mitochondrial response of Arabidopsis to flooding stress (Meng, et al. 2020). Associated with stress response (Ng, et al. 2013).
EIP10	(191 VVWQQQQ,199 GAAPPPP, 200 CCLMMMM,204 KHHKKKK, 205_AAESSSS,541_GEEGGGG, 548 TVITTTT)	-
LOC_Os03g14710.1	(381 GVAGGGG,425 QQPPPPP)	-
LOC_Os01g67720.1	(3_SLTLLLL,4_SGSGGGG, 61_VLFFFFF)	ABC1-like, ABC genes play roles in salt response in plants (Saha, et al. 2015).
LOC_Os12g24550.1	(213_VSLSSSS,421_KRKRRRR, 510_RILRRRR)	-
LOC_Os03g50390.1	(354 RKRKKKK,546 MILMMMM, 697_KRRKKKK,963_ATTAAAA, 1106_KLAKKKK)	-
protein phosphorylation | ATP binding (GO term)		
SMG2	(70 SATAAAA,131 ASTAAAA, 238_TSRRRRR,285_LLQQQQQ, 562_SSNNNNN,756_TASAAAA)	-
ATP binding | ribonucleoside triphosphate phosphatase activity (GO term)		
LOC_Os04g21110.1	GS1: (34_TIITTTT,118_NQHHHHH, 248 VIFVVVV,794 VMIVVVV) GS4: (34_TIITTTT,118_NQHHHHH, 716_NDDNNNN)	-
transferase activity, transferring phosphorus-containing groups (GO term)		
LOC_Os07g12530.1	(158 STAAAAA,395 NNDDDDD)	-
LOC_Os06g48770.1	(140_DNNDDDD,255_SAASSSS,	-

	491_FALFFFF)	
ATP binding		
LOC_Os09g36270.1	(224_HRQQQQQ,469_LLLQQQQ)	-
FTSH8	(108_TSPPPPP,407_FIFIIII, 501_VIMMMMM,532_RKRKKKK, 554_PQPQQQQ)	Plant filamentation temperature-sensitive H (FtsH) proteins was tightly associated with salt stress in *Medicago sativa* L. (Li, Zhu, et al. 2024).
DGK5	(67_APVPPPP,100_TGSGGGG)	Overexpression of *DGK5* enhanced plant susceptibility to salt stress (Li, Yao, et al. 2024).
LOC_Os02g16660.1	(12_KRRKKKK,57_YNNYYYY, 65_IVFFFFF,177_HYYHHHH)	-
TOP2	(739_SGGSSSS,760_SKSNNNN, 807_QLVQQQQ,1044_RSRAAAA)	Play roles in salt response in in Pea and Transgenic Tobacco Plants (Hettiarachchi, et al. 2005).
LOC_Os09g27420.1	(3_IATTTTT,188_NDNDDDD, 390_HHQQQQQ,534_SAASSSS, 624_TASAAAA)	-
MCCB	(353_ATITTTT,361_KRKRRRR)	-
LOC_Os05g41750.1	(27_KEEKKKK,123_AVGGGGG, 233_LMLMMMM)	-
ribonucleoside triphosphate phosphatase activity (GO term)		
LOC_Os04g52690.3	(243_ATTAAAA,328_VLIVVVV, 377_VILVVVV)	-
STAR1	(197_KENKKKK,216_VIVIIII)	Bind to ASR5 protein which play roles in stress response (Arenhart, et al. 2014; Arenhart, et al. 2016).
ABCF6	(159_LLVVVVV,557_NKEKKKK)	Member of ABC family, ABC genes play roles in salt response in plants (Saha, et al. 2015).
LOC_Os06g01980.1	(119_LPLPPPP,239_NSDSSSS, 244_PLFPPPP)	-
ABCI14	(69_SNGNNNN,92_TSTSSSS, 122_GIAVVVV)	Member of ABC family, ABC genes play roles in salt response in plants (Saha, et al. 2015).
LOC_Os03g15810.1	(47_TAPTTTT,147_IVVIIII, 222_RQQRRRR,223_IITTTTT)	-
phosphatidylinositol signaling system (KEGG pathway)		
DGK5	(67_APVPPPP,100_TGSGGGG)	Core enzyme in plant polyphosphoinositide metabolism responsing to osmotic stress (Munnik and Vermeer 2010).
PLC2	(230_RDHKKKK,339_NKNKKKK)	Core enzyme in plant polyphosphoinositide metabolism responsing to osmotic stress (Munnik and Vermeer 2010).
LOC_Os03g48300.1	(65_FYYFFFF,407_DGDGGGG, 617_EKDKKKK,835_KRRKKKK)	Downstream gene involved in InsP_6 synthesis (Wang, Cui, et al. 2024).

* Content within each bracket indicate a convergence event. The number before the underline indicates the site position in the protein sequence alignment. The seven amino acids following the underline denote: (1-3) the amino acid states of the three mangrove species; (4-6) the amino acid states of the three non-mangrove species; (7) the amino acid states of the outgroup.

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